C Programming
Lab E

QUESTION ONE: Compiling your first C program

Using vi or vim type in the following C program and call the file `labE.c`:

```
#include<stdio.h>
int main() {
   puts("Hello World\n\n");
   return 0;
}
```

Now compile that file by typing the following command at the command-line:

```
gcc labE.c
```

If you see no errors, then run the program by typing at the command-line:

```
./a.out
```

If there were errors, then look at the errors and fix them and try to compile and run again.

QUESTION TWO: Edit your first C program

Modify the file `labE.c` in the following way:

- Create an integer variable named count and initialize it to 5.
- Using a for loop, print out the Hello World, count times.

Compile and run the program.

QUESTION THREE: Write the fast fibonacci function

Write a C program fibonacci.c that performs the following:

- Read an integer n as input from stdin using scanf
- Define and call a function fib(n) that calculates and returns the nth number in the fibonacci sequence.
- This is the sequence such that F(0) = 0, F(1) = 1, and F(n) = F(n-1) + F(n-2). Translating this mathematical definition into code using recursion gives an extremely slow program.
- Instead, implement your function fib(n) so it has two temporary variables `a` and `b`. Set `a` to be F(0) and `b` to be F(1). Then perform `n` times the following: the new value of `a` is the old value of `b`, the new value of `b` is the sum of the old values of `a` and `b`.
- Finally, return `a` -- it will hold F(n).
- In the main function, print the return value of fib(n) to the user.

Compile and run the program.

You have completed your lab.

TA TUTORAIL INSTRUCTIONS

Do each part of the question waiting for the class to complete the step before going on to the next step. Give the students some time to try to solve part(s) of each question in pairs before revealing the solution. More importantly, explain *why* you're writing what you're writing.
Make sure you do everything from the mimi command line using vi or vim.

If there is time cover these extra topics:

- Show the student how to manage C language compile time errors from the command-line
    - Redirect errors to a text file (in Bash use `2>`).
    - Use the vi or vim **colon** command to jump directly to an error line number.

- Show the students how to compile their programs using a name other than a.out.

- Use any remaining time to answer general questions about the material in this lab.